

COLORADO SPRINGS, CO, USA

WMO: 724660

Lat: 38.810N

Lon: 104.688W

Elev: 6181

StdP: 11.70

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93037

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.3	7.4	-9.3	4.2	19.4	-5.2	5.2	25.9	32.3	34.8	28.5	35.6	6.4	150	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	91.1	58.7	88.7	58.4	86.1	58.3	63.7	78.4	62.5	77.5	61.6	76.7	12.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	97.0	66.6	58.4	92.0	66.0	57.0	87.3	65.7	32.1	78.2	31.2	77.3	30.4	76.7	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
28.4	25.0	21.3	-6.5	95.5	5.9	2.2	-10.7	97.2	-14.2	98.5	-17.5	99.7	-21.9	101.4		
			-7.2	66.5	5.7	1.5	-11.3	67.6	-14.6	68.4	-17.8	69.3	-22.0	70.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	31.6	32.8	40.5	46.8	56.1	66.7	71.8	69.5	62.6	50.1	39.7	31.2	
	DBStd	16.91	10.26	10.64	9.89	9.23	8.13	6.64	4.73	4.61	7.51	9.31	9.94	10.41	
	HDD50	2613	573	485	317	165	38	1	0	0	10	112	327	585	
	HDD65	6013	1037	902	759	546	289	59	8	15	126	463	760	1049	
	CDD50	2626	1	3	23	69	227	502	675	605	388	115	17	1	
	CDD65	551	0	0	0	0	13	110	217	155	54	2	0	0	
	CDH74	6989	0	0	3	28	318	1578	2476	1655	836	94	1	0	
	CDH80	2562	0	0	0	1	65	617	1065	576	231	7	0	0	
Wind	WSAvg	9.5	8.7	9.6	10.4	11.7	10.6	10.1	8.8	8.4	8.9	9.3	8.8	8.7	
Precipitation	PrecAvg	17.6	0.4	0.4	1.2	1.6	2.2	2.4	2.9	3.5	0.9	0.8	0.5	0.5	
	PrecMax	27.6	1.2	2.4	2.4	7.5	5.0	6.0	5.3	7.0	3.0	5.0	2.0	1.1	
	PrecMin	12.4	0.0	0.0	0.3	0.0	0.6	0.1	0.7	0.4	0.0	0.0	0.0	0.0	
	PrecStd	3.7	0.3	0.5	0.7	1.6	1.3	1.5	1.5	1.7	0.7	1.0	0.6	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	66.5	72.7	77.6	86.0	94.8	94.4	91.5	89.4	80.5	72.1	62.6	
		MCWB	41.8	42.4	45.7	48.8	54.3	56.6	60.2	59.8	58.0	52.4	45.9	41.4	
	2%	DB	58.5	60.6	68.1	73.3	81.4	90.2	91.6	88.6	85.4	76.8	67.1	57.9	
		MCWB	39.2	40.2	43.5	47.3	52.4	56.6	59.4	58.9	56.1	50.3	43.9	39.2	
	5%	DB	53.7	56.3	63.8	69.6	77.6	87.0	89.3	86.1	82.6	72.7	62.7	53.5	
		MCWB	37.1	38.3	42.0	45.7	51.3	56.1	59.5	59.4	55.6	49.0	42.3	37.2	
	10%	DB	49.3	51.2	59.1	65.4	73.4	83.3	86.6	83.3	79.3	68.1	57.7	48.9	
		MCWB	34.8	35.8	40.3	44.4	50.6	55.9	59.6	59.3	54.9	47.3	40.3	34.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.6	43.6	46.9	51.3	58.5	63.3	65.5	65.1	62.3	55.3	47.1	42.3
			MCDB	61.5	63.6	68.0	68.9	74.4	78.6	78.5	78.7	78.5	69.8	67.2	60.7
2%		WB	39.6	40.9	44.8	49.0	56.1	61.2	64.1	63.5	59.9	53.0	44.8	39.7	
		MCDB	56.7	59.1	64.8	67.2	71.8	76.8	78.4	78.3	75.8	69.5	64.9	56.3	
5%		WB	37.4	38.8	43.0	47.3	54.2	59.8	62.9	62.4	58.4	50.8	42.9	37.4	
		MCDB	52.7	55.1	61.6	65.5	70.0	76.8	77.9	77.1	74.3	67.8	61.1	52.6	
10%		WB	35.2	36.4	41.0	45.6	52.6	58.6	62.0	61.4	57.0	48.7	40.8	35.1	
		MCDB	48.1	50.5	57.7	63.2	68.1	75.8	77.3	75.9	72.8	65.3	56.7	48.4	
Mean Daily Temperature Range	5% DB	MDBR	24.3	24.2	25.5	24.5	24.7	26.6	26.0	24.6	25.9	25.6	24.9	23.8	
		MCDBR	31.0	31.7	32.1	32.4	31.8	32.5	30.8	29.2	31.1	32.5	32.5	30.4	
		MCWBR	16.7	16.3	14.7	14.0	11.8	10.5	9.2	9.0	10.6	13.8	15.6	16.4	
	5% WB	MCDBR	30.1	30.7	30.4	29.9	26.8	27.0	24.8	24.0	26.1	28.2	30.5	29.7	
		MCWBR	16.6	16.4	14.4	13.5	11.0	10.4	8.8	9.0	9.9	12.8	15.2	16.4	
Clear-Sky Solar Irradiance	taub	0.232	0.236	0.257	0.283	0.302	0.315	0.343	0.331	0.296	0.262	0.241	0.234		
	taud	2.629	2.648	2.584	2.520	2.528	2.510	2.479	2.503	2.574	2.652	2.631	2.620		
	Ebn at Noon	312	323	322	316	310	304	294	297	304	308	305	303		
	Edh at Noon	21	24	28	32	33	33	34	33	29	23	21	20		
All-Sky Solar Radiation	RadAvg	863	1162	1559	1886	2088	2326	2199	2000	1725	1296	947	762		
	RadStd	44	74	114	116	156	105	92	68	88	76	53	40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.01	N/A	N/A	+1.04	N/A	N/A	N/A	-252	+221	+110
Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+40

Nomenclature: See separate page